NextGen EV Powertrain & Battery Management System

AUTOSAR Classic Platform 4.4 High-Level Software Design (HLD)

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This register tracks engineering revisions, baseline releases, and engineering sign-offs in accordance with ASPICE SWE.1 and ISO 26262 functional safety processes.

Rev	Date	Author	Engineering Scope & Changes	Status
1.0	2025-04-12	H. Schmidt	Initial architectural baseline and component decomposition	Approved
2.0	2025-10-18	M. Tanaka	Added ThermalManagement SWC and cooling loop controls	Approved
2.3	2026-03-05	D. Miller	High-voltage contactor state machine and precharge logic	Approved
3.0	2026-06-20	E. Vance	Integrated CCS/NACS ChargingControl and ISO 15118 stack	Approved
3.4	2026-10-15	A. Kowalski	Updated port matrix, cell balancing, and safety watchdog SWC	Released

Engineering Approvals Sign-off

Role	Name	Department	Signature Date
Lead Systems Architect	Dr. Marcus Vance	E/E Architecture Group	2026-10-16
Functional Safety Manager	Elena Rostova	Automotive Safety & Compliance	2026-10-17
Software Project Lead	Rajesh Nair	Powertrain Software Delivery	2026-10-18

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1. Executive Summary & Vehicle E/E Architecture Context

This High-Level Design document establishes the formal software architecture for the Battery Management System (BMS) and Powertrain Coordination Controller within the Platform-X 800V electric vehicle platform. The system operates in a distributed automotive environment connecting through dual CAN FD buses and an Automotive Ethernet backbone.

The primary mission of the BMS software is to maintain battery pack safety under all operational conditions, accurately estimate State of Charge (SOC) and State of Health (SOH), coordinate thermal cooling and heating circuits, control high-voltage contactor precharge and isolation, and support high-power DC fast charging up to 350 kW.

Key Architectural Characteristics:

• **Layered AUTOSAR Architecture:** Full separation between Application Software Components (ASW), Runtime Environment (RTE), and Basic Software (BSW).

• **Dual Microcontroller Cores:** Core 0 runs safety-critical voltage monitoring and precharge tasks; Core 1 executes thermal control and state estimation.

• **ISO 26262 ASIL-D Compliance:** Redundant voltage and current measurement plausibility with hardware-enforced emergency shut-off paths.

• **Real-time Determinism:** Cyclic task execution with 1 ms, 5 ms, 10 ms, and 100 ms periodicities scheduled by the AUTOSAR OSEK/VDX OS.

2. Functional Safety Concept & ISO 26262 ASIL Decomposition

In accordance with ISO 26262:2018 Road Vehicles — Functional Safety, a Hazard Analysis and Risk Assessment (HARA) identified critical top-level vehicle hazards relating to high-voltage battery failure modes. The following safety goals are assigned to the software architecture:

Safety Goal ID	Safety Goal Description	ASIL	FTTI	Allocated SWC
SG-01	Prevent battery cell overcharge leading to thermal runaway	ASIL-D	50 ms	BatteryManagement
SG-02	Prevent unintended closing of main high-voltage contactors	ASIL-D	20 ms	PowerControl
SG-03	Ensure high-voltage bus discharge upon crash detection	ASIL-B	100 ms	VehicleSafetySupervisor
SG-04	Prevent pack overheating during extreme ambient DC fast charge	ASIL-B	500 ms	ThermalManagement
SG-05	Detect high-voltage isolation loss to vehicle chassis ground	ASIL-C	200 ms	PowerControl

Fault Tolerant Time Interval (FTTI) Budgeting

For SG-01, the overall FTTI is 50 ms. The software detection budget is allocated 20 ms, leaving 30 ms for hardware actuator response (opening solid-state pyrofuse and main pack contactors). The runnable Runnable_BattMonitor executes every 10 ms to guarantee double-sampling verification within the allotted budget.

3. Hardware Platform & Microcontroller Mapping

The target ECU hardware incorporates an Infineon AURIX TC397x TriCore microcontroller featuring 6 cores clocked at 300 MHz, with 16 MB embedded flash and 2.9 MB SRAM. Lockstep cores (Core 0 and Core 1) provide hardware safety redundancy for ASIL-D computation.

Core ID	Architecture Role	Software Allocation	Safety Level
Core 0 (Lockstep)	Master Safety Controller	BatteryManagement, VehicleSafetySupervisor	ASIL-D
Core 1 (Lockstep)	Actuation & Power Control	PowerControl, InverterInterface	ASIL-D
Core 2	Thermal & Auxiliary Processing	ThermalManagement, CellBalancing	ASIL-B
Core 3	Charging & Communications	ChargingControl, COM Stack, Ethernet	ASIL-B / QM
Core 4	Diagnostics & Telemetry	Diagnostics SWC, DEM, DCM, UDS Stack	QM
Core 5	Reserved / System Monitor	OS Task Timing Watchdog, Memory BIST	ASIL-D

Hardware Interfaces & Peripherals

• **Dual IsoSPI Channels:** High-speed isolated serial links connecting 12 daisy-chained Battery Cell Monitoring ICs (ADBMS6815).

• **CAN FD Controller 0:** Powertrain Backbone Bus (500 kbps arbitration / 2.0 Mbps data phase).

• **Automotive Ethernet (100BASE-T1):** Diagnostics, flash reprogramming, and vehicle cloud telemetry.

4. AUTOSAR Layered Architecture & RTE Overview

The software architecture strictly adheres to AUTOSAR Classic Platform Release 4.4.0. The application layer comprises 8 Atomic Software Components (AtomicSWC) interacting exclusively through the AUTOSAR Runtime Environment (RTE). Direct cross-component memory sharing or bypass calls are strictly prohibited.

AUTOSAR Architectural Layers:

1. **Application Software Layer (ASW):** Encapsulates automotive domain algorithms including state-of-charge calculation, contactor state sequences, and temperature regulation.

2. **Runtime Environment (RTE):** Provides standardized communication infrastructure via Sender-Receiver (S/R) and Client-Server (C/S) paradigms.

3. **Basic Software Layer (BSW):** Supplies system services (OS, EcuM, BswM), memory abstraction (NvM, MemIf, Fee), and communication drivers (CanIf, Com, PduR).

4. **Microcontroller Abstraction Layer (MCAL):** Low-level hardware drivers (Gpt, Spi, Dio, Adc, Can) interfacing with AURIX silicon.

Communication Paradigms Used in this Architecture:

• **Sender-Receiver (S/R):** Used for continuous telemetry data flow (e.g. cell voltages, module temperatures, pack current, and state-of-charge).

• **Client-Server (C/S):** Used for asynchronous service invocations (e.g. requesting contactor opening, triggering NvM write, initiating DTC reset).

5. BatteryManagement Software Component Specification

The BatteryManagement SWC represents the foundational component responsible for measuring pack voltages, estimating aggregate State of Charge (SOC), and detecting cell-level anomalies.

Component Attributes:

• **Component Type:** Application Software Component (AtomicSWC)
• **Assigned Core:** Core 0 (Lockstep)
• **ASIL Level:** ASIL-D
• **Execution Runnables:** Runnable_BattInit (Init), Runnable_BattMonitor (10 ms cyclic), Runnable_BattFault (Event-triggered).

Port Definitions for BatteryManagement SWC:

Port Name	Direction	Interface	Description	Data Type
BattVoltage_Prov	P-Port	BattVoltage_If	Provides individual and pack voltage	float32[96]
BattSOC_Prov	P-Port	BattSOC_If	Provides estimated State of Charge	float32
BattCurrent_Prov	P-Port	BattCurrent_If	Provides calibrated shunt pack current	float32
BattFault_Prov	P-Port	FaultStatus_If	Provides battery fault status register	uint32
BattTemp_Req	R-Port	TempSensor_If	Requires module temperature readings	float32[16]
CellBalancing_Req	R-Port	BalancingCmd_If	Demands cell passive balancing action	uint16

6. ThermalManagement Software Component Specification

The ThermalManagement SWC orchestrates multi-zone liquid cooling, chiller valves, and resistive pack heaters to maintain lithium-ion cell temperatures between 20°C and 35°C.

Component Attributes:

- **Component Type:** Sensor-Actuator Software Component
- **Assigned Core:** Core 2
- **ASIL Level:** ASIL-B
- **Execution Runnables:** Runnable_ThermalInit (Init), Runnable_ThermalControl (50 ms cyclic).

Port Definitions for ThermalManagement SWC:

Port Name	Direction	Interface	Description	Data Type
CoolingCmd_Prov	P-Port	CoolingCmd_If	Outputs variable cooling pump/fan command	uint8 (0-100%)
HeaterCmd_Prov	P-Port	HeaterCmd_If	Demands PTC coolant heater power level	uint16 (Watts)
TempSensor_Prov	P-Port	TempSensor_If	Publishes validated thermistor values	float32[16]
BattSOC_Req	R-Port	BattSOC_If	Requires SOC for thermal throttling limits	float32
BattFault_Req	R-Port	FaultStatus_If	Requires pack fault state to trigger E-Coo	uint32
AmbientTemp_Req	R-Port	VehicleEnv_If	Requires vehicle external temperature	float32

7. PowerControl & High-Voltage Contactor Management SWC

The PowerControl SWC governs high-voltage contactor opening, precharge resistor timing, and emergency isolation. It enforces strict timing guards to avoid contactor welding and catastrophic DC bus arcing.

Contactor State Machine Transitions:

1. **OPEN:** Main positive, main negative, and precharge contactors are de-energized.

2. **PRECHARGE:** Negative contactor closes; precharge contactor engages for 250 ms until inverter DC link reaches 95% of pack voltage.

3. **CLOSED:** Main positive contactor closes; precharge contactor opens. High-voltage path is fully energized.

4. **FAULT_TRIP:** Instantaneous contactor drop (< 15 ms) upon ASIL-D fault flag.

Port Definitions for PowerControl SWC:

Port Name	Direction	Interface	Description	Data Type
RelayCmd_Prov	P-Port	RelayCmd_If	Commands main contactor driver relays	uint8 (bitmap)
PrechargeStatus_Prov	P-Port	Precharge_If	Reports DC bus precharge completion	boolean
BattVoltage_Req	R-Port	BattVoltage_If	Requires pack voltage to verify precharge	float32[96]
BattSOC_Req	R-Port	BattSOC_If	Requires SOC to prevent under-voltage load	float32
BattFault_Prov	P-Port	FaultStatus_If	NOTE: Flagged inconsistency - should be R-Port	uint32
EStop_Req	R-Port	SafetyTrip_If	Direct hardware safety trip trigger	boolean

8. StateOfHealth & CellBalancing SWC Specification

The StateOfHealth and CellBalancing SWC executes internal resistance tracking, capacity fade degradation modeling, and autonomous passive cell balancing across cell strings.

Algorithmic Methodology:

• **Dual Extended Kalman Filter (DEKF):** Concurrently estimates cell internal state and remaining amp-hour capacity over lifecycle aging.

• **Passive Bleed Balancing:** Balances cells displaying > 15 mV delta during vehicle rest/charging states using 100 ohm bypass resistors.

Port Definitions for CellBalancing SWC:

Port Name	Direction	Interface	Description	Data Type
SOH_Metrics_Prov	P-Port	SOH_Data_If	Provides State of Health percentage and IR	float32
BalancingCmd_Prov	P-Port	BalancingCmd_If	Controls individual cell bypass switches	uint32[3]
BattVoltage_Req	R-Port	BattVoltage_If	Requires individual cell voltage values	float32[96]
BattCurrent_Req	R-Port	BattCurrent_If	Requires historical integral current (Ah)	float32
TempSensor_Req	R-Port	TempSensor_If	Requires thermal gradient across modules	float32[16]

9. ChargingControl SWC Specification (AC & DC Fast Charge)

The ChargingControl SWC coordinates vehicle EVSE communication, charge current limits, pilot signal state monitoring, and digital handshake protocols in accordance with ISO 15118 and DIN 70121.

Charging Modes Supported:

• **AC Level 2 (Type 2 / J1772):** On-board charger up to 22 kW three-phase with automated lock actuator control.

• **DC Fast Charging (CCS2 / NACS):** Off-board charging up to 800V / 500A (350 kW) with continuous temperature monitoring.

Port Definitions for ChargingControl SWC:

Port Name	Direction	Interface	Description	Data Type
ChargeTarget_Prov	P-Port	ChargeTarget_If	Publishes allowable EVSE voltage & current	uint16[2]
InletLock_Prov	P-Port	LockActuator_If	Controls charging flap/lock solenoid	uint8
BattSOC_Req	R-Port	BattSOC_If	Requires pack SOC to apply charging taper	float32
TempSensor_Req	R-Port	TempSensor_If	Monitors inlet and cell temperatures	float32[16]
EVSE_Comm_Req	R-Port	ISO15118_If	Receives EVSE status via HomePlug Green PHY	uint8[64]

10. InverterInterface & Traction Drive Coordination SWC

The InverterInterface SWC acts as the high-speed gateway between the Battery Management System and the dual-motor front/rear silicon-carbide (SiC) traction inverters.

Key Responsibilities:

• **Dynamic Power Limit Calculation:** Computes maximum discharge (acceleration) and regen (braking) power limits updated every 10 ms.

• **DC Bus Active Discharge:** In the event of crash or service shutdown, commands inverters to bleed high-voltage bus below 60V within 2.0 seconds.

Port Definitions for InverterInterface SWC:

Port Name	Direction	Interface	Description	Data Type
MaxDischarge_Prov	P-Port	PowerLimit_If	Provides maximum allowed discharge kW	uint16
MaxRegen_Prov	P-Port	PowerLimit_If	Provides maximum allowed regenerative kW	uint16
DischargeCmd_Prov	P-Port	ActiveBleed_If	Triggers active inverter discharge gate drive	boolean
InverterStatus_Req	R-Port	InvStatus_If	Monitors inverter DC link voltage & temp	uint32
BattVoltage_Req	R-Port	BattVoltage_If	Requires pack voltage to compute sag limits	uint32[96]

11. VehicleSafetySupervisor & Watchdog SWC (ASIL-D)

The VehicleSafetySupervisor SWC runs on the primary lockstep core (Core 0) and performs independent, diversified plausibility checking of all voltage, current, and temperature inputs to prevent single-point failures.

Safety Mechanisms Implemented:

• **E2E Plausibility Check:** Compares analog hardware voltage taps with serial IsoSPI digital stream.

• **Over-Current Hardware Interruption:** Coordinates 1200A pyrofuse ignition in case of hard short circuit within 2 ms.

• **Program Flow Monitoring:** Verifies that all ASIL-D cyclic runnables execute in correct sequence without deadline overrun.

Port Definitions for VehicleSafetySupervisor SWC:

Port Name	Direction	Interface	Description	Data Type
SafetyTrip_Prov	P-Port	SafetyTrip_If	Outputs definitive system-level trip signal	boolean
WatchdogTrigger_Prov	P-Port	WdgM_If	Services external hardware watchdog IC	uint8
BattVoltage_Req	R-Port	BattVoltage_If	Requires raw voltages for cross-validation	float32[96]
BattCurrent_Req	R-Port	BattCurrent_If	Requires raw current for over-current check	float32
CrashSensor_Req	R-Port	AirbagSignal_If	Direct hardwire input from airbag crash ECU	boolean

12. Diagnostics & Fault Management SWC (DEM / DCM / UDS)

The Diagnostics SWC manages Diagnostic Trouble Codes (DTC), freeze frame snapshot recording, and UDS ISO 14229 client service handling over CAN FD and Ethernet DoIP.

Diagnostic Trouble Codes (DTC) Registry:

DTC Code	Hex ID	Severity	Description	Failsafe Action
U0100	0xC10008	Critical	Lost Communication with Inverter Gateway	Open contactors gracefully
P0A80	0x0A8000	High	Replace Hybrid/EV Battery Pack Degradation	Limit power to 50 kW
P0AFA	0x0AFA16	Critical	Battery Voltage Sensor Circuit Plausibility	Prevent driving engagement
P0B29	0x0B2900	Medium	Hybrid Battery Pack Cooling Fan Performance	Derate charge power
P0C17	0x0C1717	High	Drive Motor Position Sensor Calibration Error	Enter Limp-Home mode

Port Definitions for Diagnostics SWC:

Port Name	Direction	Interface	Description	Data Type
DTC_Event_Prov	P-Port	Dem_Event_If	Reports fault status flags to DEM stack	uint32
FreezeFrame_Prov	P-Port	Dcm_Snapshot_If	Supplies cell snapshot buffer to DCM	uint8[128]
BattFault_Req	R-Port	FaultStatus_If	Listens for active component faults	uint32

13. Port and Interface Definition Matrix

This master matrix consolidates all AUTOSAR PortInterfaces across all 8 software components to verify complete communication closure and interface binding integrity.

Interface Name	Interface Type	Provider Component	Requirer Component(s)	Cycle
BattVoltage_If	Sender-Receiver	BatteryManagement	PowerControl, Inverter, Safety	10 ms
BattSOC_If	Sender-Receiver	BatteryManagement	PowerControl, Thermal, Charging	100 ms
BattCurrent_If	Sender-Receiver	BatteryManagement	CellBalancing, SafetySupervisor	5 ms
FaultStatus_If	Sender-Receiver	BatteryManagement	Thermal, Diagnostics, PowerControl	Event
TempSensor_If	Sender-Receiver	ThermalManagement	BatteryManagement, CellBalancing	50 ms
CoolingCmd_If	Sender-Receiver	ThermalManagement	Thermal Actuator Drivers	100 ms
RelayCmd_If	Sender-Receiver	PowerControl	MCAL Dio Driver	10 ms
SafetyTrip_If	Sender-Receiver	VehicleSafetySupervisor	PowerControl (Emergency Open)	1 ms
Dem_Event_If	Client-Server	Diagnostics	BSW DEM Service	Event

Interface Integrity Rules:

1. Every P-Port must have at least one valid matching R-Port in the composition.

2. Data types, arrays lengths, and unit scaling must match precisely between provider and requirer.

14. Data Dictionary & Global Signal Catalogue

The Data Dictionary standardizes the naming, base type, physical minimum, maximum, scaling factor, and initial value for all signals exchanged over the RTE.

Signal Name	Data Type	Min	Max	Resolution	Unit	Init Val
CellVoltage_mV	uint16[96]	1500	4500	1.0	mV	3700
PackCurrent_A	float32	-800.0	800.0	0.01	A	0.0
PackSOC_Pct	float32	0.0	100.0	0.1	%	50.0
MaxTemp_DegC	float32	-40.0	125.0	0.25	°C	25.0
ContactorState	uint8	0	4	1	Enum	0 (Open)
Isolation_kOhm	uint16	0	65535	1.0	kΩ	10000
PyrofuseState	boolean	0	1	1	Flag	0 (Intact)
ChargeTarget_kW	uint16	0	350	1.0	kW	0

Signal Integrity Conventions:

• All voltages are represented in millivolts (mV) to maintain integer precision on Core 0.

• Pack current incorporates an active zero-point offset calibrated upon every contactor open state.

15. Runnable Entities & OS Task Scheduling Allocation

Runnables represent atomic units of execution invoked by the AUTOSAR OS scheduler via RTE event triggers. Execution timing budgets must never exceed 70% of task period.

Runnable Entity	Host SWC	Trigger Event	Period	Budget	Target Core
Runnable_SafetyAudit	SafetySupervisor	TimingEvent	1 ms	120 µs	Core 0
Runnable_CurrentSample	BatteryManagement	TimingEvent	5 ms	350 µs	Core 0
Runnable_BattMonitor	BatteryManagement	TimingEvent	10 ms	850 µs	Core 0
Runnable_ContactorSeq	PowerControl	TimingEvent	10 ms	400 µs	Core 1
Runnable_ThermalCtrl	ThermalManagement	TimingEvent	50 ms	1.2 ms	Core 2
Runnable_SOC_Estimate	BatteryManagement	TimingEvent	100 ms	4.5 ms	Core 0
Runnable_CellBalancing	CellBalancing	TimingEvent	100 ms	1.8 ms	Core 2
Runnable_DcmService	Diagnostics	DataReceived	Event	2.0 ms	Core 4

Scheduling Hierarchy:

• Highest priority: Task_1ms (Priority 64, Non-preemptible safety loop).

• Medium priority: Task_10ms (Priority 48, Preemptible control loops).

• Background: Task_100ms (Priority 16, Telemetry and diagnostics).

16. In-Vehicle Communication Matrix (CAN FD & Ethernet)

The BMS ECU interfaces with the vehicle powertrain backbone using ISO 11898-1:2015 CAN FD (500 kbps arbitration / 2.0 Mbps payload data phase) and Automotive Ethernet SOME/IP.

Frame Name	CAN ID	Payload	Cycle	Transmitter	Receiver(s)
BMS_Status_01	0x110	32 bytes	10 ms	BMS ECU	Inverter, VCU, Gateway
BMS_PowerLimits	0x112	16 bytes	10 ms	BMS ECU	Traction Inverter
BMS_CellVoltages	0x250	64 bytes	50 ms	BMS ECU	Instrument Cluster, VCU
VCU_TorqueDemand	0x090	16 bytes	10 ms	VCU	BMS, Inverter
EVSE_ChargeParam	0x380	48 bytes	100 ms	OBC / EVSE	BMS ChargingControl

Automotive Ethernet SOME/IP Services:

• **Service ID 0x4010 (Pack Telemetry):** High-bandwidth stream delivering 96 individual cell voltages and 16 thermistor streams to onboard telemetry unit.

• **Service ID 0x4020 (Diagnostic Log):** UDS over DoIP (ISO 13400) diagnostic tunnel.

17. Non-Volatile Memory (NvM) Block Layout

Persistent operational parameters, calibration tables, and lifetime history are managed through the AUTOSAR NvM service layer with redundant CRC32 validation and wear-leveling.

NvM Block	Block ID	Size	Management Type	CRC Type	Description
NvM_BMS_Calib	0x0001	256 B	NATIVE	CRC32	Shunt resistance & ADC voltage offsets
NvM_SOH_History	0x0002	512 B	REDUNDANT	CRC32	Lifetime Ah throughput and cycle count
NvM_DTC_Store	0x0003	1024 B	DATASET	CRC16	Active and historical freeze frames
NvM_CellBalances	0x0004	128 B	NATIVE	CRC16	Accumulated bypass bleed timestamps

NVRAM Recovery Strategy:

Upon CRC mismatch during startup EcuM initialization, the NvM restores the default ROM calibration block and asserts a diagnostic event (DTC P0607).

18. End-to-End (E2E) Communication Protection Concept

To satisfy ASIL-D functional safety requirements against communication transit faults (masquerading, insertion, corruption, and loss), critical frames implement AUTOSAR E2E Profile 01.

E2E Header Structure:

- **Byte 0:** CRC8 polynomial 0x1D (SAE J1850 standard).
- **Byte 1:** 4-bit Alive Counter (0–14) + 4-bit Data ID nibble.
- **Bytes 2–N:** Protected application payload.

Fault Tolerant Rejection Thresholds:

- **Alive Counter Inconsistency:** > 2 consecutive missed increments triggers fault.
- **CRC Verification Failure:** 1 invalid checksum causes immediate frame discard.
- **Timeout Guard:** If no valid frame is received within 30 ms (3x cycle), failsafe default values are applied.

19. Vehicle Mode Management & Power State Machine

The Vehicle Mode Manager (VMM) dictates overall BMS operational states in response to ignition key inputs, crash signals, and EVSE plug insertions.

State Name	HV Contactors	Active SWCs	CAN Bus Activity	Target Power
STANDBY	OPEN	SafetySupervisor (100 ms)	Awake / Listening	< 0.5 W
PRECHARGE	PRECHARGE ON	PowerControl, BatteryManagement	High-speed active	25 W
DRIVE	CLOSED	All SWCs active	Full CAN FD & Ethernet	45 W
CHARGING	CLOSED	ChargingControl, ThermalManagement	Active EVSE loop	35 W
EMERGENCY	FORCE OPEN	SafetySupervisor, Diagnostics	Emergency broadcasts	10 W

Transition Constraints:

• Transition from STANDBY to DRIVE requires successful insulation resistance check (> 500 kΩ).

• Crash signal detection transitions the system directly to EMERGENCY state within 5 ms, bypassing all ramp-down timers.

20. Cross-Component Sequences & Interaction Flows

This section documents the end-to-end execution sequence for primary vehicle power-up and high-voltage bus precharging.

Sequence 1: Normal Vehicle Power-On Precharge Flow

1. Vehicle Control Unit transmits VCU_DriveEnable command over CAN FD.
2. **VehicleSafetySupervisor** confirms no pending ASIL-D fault flags in **FaultStatus_If**.
3. **PowerControl** closes the Main Negative Contactor via RelayCmd_Prov.
4. **PowerControl** engages the Precharge Contactor; current flows through the 47 ohm precharge resistor.
5. **BatteryManagement** samples cell string voltage while **InverterInterface** monitors DC bus rise.
6. When V_Bus >= 0.95 * V_Pack, **PowerControl** closes Main Positive Contactor and opens Precharge Contactor.
7. **PowerControl** asserts PrechargeStatus_Prov = TRUE; drive authorization granted.

21. Intentional Architectural Inconsistencies & Known Issues

Notice for Evaluators & Automated Verification Engines: This document contains intentionally introduced architectural inconsistencies to benchmark the AutoHLD consistency auditing and error detection pipeline.

Issue ID	Category	Target Component	Page Ref	Discrepancy Description
INC-01	Port Direction Mismatch	PowerControl	Page 10	BattFault_Prov declared as P-Port; should be R-Port.
INC-02	Undefined Interface	PowerControl	Page 10	SafetyTrip_If interface referenced but direction ambiguous.
INC-03	Data Type Mismatch	BatteryManagement	Page 8	BattVoltage_Prov uses float32[96] while Table 14 lists uint16[96].
INC-04	Missing Description	CellBalancing	Page 11	BalancingCmd_Prov lacks explicit timeout safety fallback.
INC-05	Cross-Reference Conflict	ThermalManagement	Page 9	AmbientTemp_Req interface VehicleEnv_If missing provider.

Automated Review Expectation:

AutoHLD's consistency rule checker should detect these five specific discrepancies during automated review and categorize INC-01 and INC-03 as high-priority Warnings.

22. Integration Guidelines & RTE Directives

When generating the AUTOSAR RTE via DaVinci Developer or EB tresos, enable Rte_Read and Rte_Write macro inlining for 1 ms tasks to minimize stack overhead.

23. Verification & Validation Criteria

- **Model-in-the-Loop (MIL):** 100% statement and MC/DC coverage for ASIL-D runnables.
- **Hardware-in-the-Loop (HIL):** Automated testing on dSPACE SCALEXIO with high-voltage battery emulator.

24. Glossary of Automotive Terms

Term	Definition
AUTOSAR	AUTOmotive Open System ARchitecture standard for automotive ECUs.
SWC	Software Component — atomic functional unit in the AUTOSAR ASW.
RTE	Runtime Environment — software middleware decoupling ASW from BSW.
P-Port	Provided Port — communication endpoint where an SWC outputs data or services.
R-Port	Required Port — communication endpoint where an SWC inputs data or services.
ASIL-D	Automotive Safety Integrity Level D — highest rigor safety classification in ISO 26262.
FTTI	Fault Tolerant Time Interval — duration between fault occurrence and hazardous event.
UDS	Unified Diagnostic Services (ISO 14229) — standard automotive diagnostic protocol.